

As many as 85% of infants born with congenital heart disease can be expected to survive into adulthood.¹ In the United States, the total number of adults with congenital heart disease is increasing at a rate of about 5% per year. In the current era, more adults have congenital heart disease than children. This number includes both unrepaired and surgically corrected patients.² The population of patients with congenital heart disease who reach adult life grows each year because of advances in interventional and noninvasive cardiology, cardiothoracic surgery, and intensive care. As such, a growing number of adult patients with congenital heart disease are likely to present to the general cardiologist as opposed to a highly specialized adult congenital heart disease clinic. The American College of Cardiology Task Force 1 on congenital heart disease estimated that at least 10% of patients with congenital heart disease are diagnosed as adults. Furthermore, the consensus opinion of this committee was that the number of adults with undiagnosed congenital heart disease is increasing due to the growing immigrant population.¹ Thus it is important that all cardiologists possess a sound understanding of the hemodynamics associated with the common congenital heart disease lesions as this provides valuable insight into the pathophysiology of these conditions.

Our understanding of the physiology of congenital heart disease in humans was largely theoretical until the 1940s when Dexter et al.³ published the first manuscript that described the use of right heart catheterization to directly measure intracardiac pressure and oxygenation saturations in patients with congenital heart disease. Since then, the complete assessment of congenital heart disease has evolved to include echocardiography and magnetic resonance imaging, making reliance on diagnostic catheterization data a less frequent occurrence.

This chapter will focus on lesions more commonly seen in an adult cardiologist's practice, including atrial septal defect (ASD), ventricular septal defect (VSD), coarctation of the aorta, and Ebstein anomaly of the tricuspid valve. It will also focus on the postoperative, surgically palliated patient with tetralogy of Fallot, the patient with peripheral pulmonary artery stenosis, and the patient with Eisenmenger syndrome.

ATRIAL SEPTAL DEFECT

ASDs are the second most common congenital heart defect, following VSDs. This communication results in a left-to-right shunt, leading to a volume load to the right atrium, right ventricle, and pulmonary arteries. The volume load leads to distention of the right atrial and right ventricular myocardium, which, in turn, can promote diastolic dysfunction and lead to arrhythmias. Additionally, the increase in pulmonary blood flow can cause shear stress on the pulmonary endothelium and lead to the development of vascular disease with resulting pulmonary and right ventricular hypertension.

During fetal circulation, the interaction of the inferior/leftward primum and superior/rightward secundum septa forms the foramen ovale, which functions as a valve allowing right-to-left shunting of oxygenated placental blood into the left heart to be ejected preferentially to the cerebral and upper extremity circulation. Following birth, a progressive increase in pulmonary blood flow leads to an increase in pulmonary venous return, an increased left atrial pressure, and subsequent closure of the foramen ovale. The overall incidence of a patent foramen ovale has been estimated to be as high as 27% but has been reported to decline with increasing age.⁴ A patent foramen ovale rarely results in significant clinical sequelae and is not associated with hemodynamic abnormalities. The notable (and often debated) clinical exception is the case of a paradoxical embolus, presumed to occur via a patent foramen ovale allowing transient right-to-left shunting.

Four types of interatrial communications have important clinical sequelae: (1) primum ASD, (2) secundum ASD, (3) sinus venosus defect, and (4) coronary sinus septal defect.² The majority of cases occur spontaneously; however, reports of familial cases exist.⁵

After patent foramen ovale, the secundum ASD is the most common type of interatrial communication. It comprises up to 75% of ASDs. The interatrial communication may be due to a single hole or multiple fenestrations in the septum primum. In rare cases the secundum ASD may result from an incomplete septum secundum. The pathophysiologic manifestation of a secundum ASD is a left-to-right shunt across the atrial septum, with resultant volume overload of the right heart. Campbell's⁶ natural history studies have shown that, with time, right-sided heart failure and pulmonary hypertension develop and lead to early mortality.

A sinus venosus defect occurs when the tissue between either the vena cava, the right atrium, or the pulmonary veins fails to incorporate properly. It comprises up to 10% of ASDs.² The most frequently encountered sinus venosus defect involves a communication between the superior vena cava (SVC)/right atrial junction and the right upper pulmonary vein.^{7,8} Less frequently, these defects involve other right-sided pulmonary veins and the inferior vena cava/right

*With contributions from the first edition by Rajan Patel, MD, and D. Scott Lim, MD.

atrial junction. Patients with sinus venosus defects frequently have a partial anomalous pulmonary venous return, with the right upper pulmonary vein draining to the SVC.

The primum ASD is the third most common type of interatrial communication after patent foramen ovale and secundum ASD. It comprises up to 15% of ASDs.² The defect is due to maldevelopment of the endocardial cushions and is associated with a cleft in the anterior mitral leaflet and mitral regurgitation.

The least frequently encountered ASD is the coronary sinus septal defect. The tissue that constitutes the wall between the coronary sinus and the left atrium is either completely absent or only partially developed. Therefore the left atrium and right atrium are connected via the coronary sinus.

HEMODYNAMICS

The volume of shunt across an interatrial communication is a function of the size of the defect and the relative compliance of the right and left ventricles, which was first suggested by Dexter in 1956.⁹ These variables change throughout life with some defects becoming smaller during periods of myocardial growth in addition to dynamic changes in ventricular compliance during development.

As a neonate, right ventricular compliance is low, reflecting its relatively high workload during fetal development. At this stage, there tends to be a relatively small left-to-right shunt for a given defect size. As the neonate progresses to infancy and pulmonary vascular resistance reduces, the pressure load of the right ventricle declines, resulting in a smaller and more compliant chamber. As the right ventricle continues to become more compliant relative to the left ventricle, flow across an atrial defect will increase throughout childhood and early adolescence. This results in a volume load to the right atrium, right ventricle, and pulmonary arteries. With a significant shunt, this will be demonstrated by dilation of those chambers and pulmonary vessels. Given a significant shunt and adequate time (typically decades) endothelial damage of the pulmonary vasculature and myocardial maladaptation ensue, leading to pulmonary hypertension, arrhythmias, and right ventricular systolic and diastolic dysfunction. If right ventricular diastolic function continues to decline, the atrial shunt may reverse direction from right to left, resulting in systemic hypoxemia.¹⁰

Cardiac catheterization is performed for hemodynamic assessment and potential closure of secundum ASDs. Right-sided heart catheterization demonstrates an increase in saturations from the SVC into the right atrium, with a significant shunt generally having a pulmonary flow (Q_p) to systemic flow (Q_s) ratio (or Q_p/Q_s) ≥ 1.4 . Right atrial pressures tend to be normal but may be elevated if there is right ventricular diastolic dysfunction (Fig. 13.1). Additionally, the atrial tracing may demonstrate a dominant v wave if the atrial defect is large. Right ventricular pressures tend to be normal unless pulmonary hypertension or diastolic dysfunction has developed because of a long-standing shunt.

Closure of an ASD when there is left-sided diastolic dysfunction, common in later decades of life or in those with significant coronary artery disease, should be undertaken with care as the atrial defect acts to volume-unload the left ventricle. With closure of the defect, the increased volume load may elevate left ventricular end-diastolic pressure and subsequently elevate left atrial and pulmonary venous pressure, potentially causing pulmonary congestion.

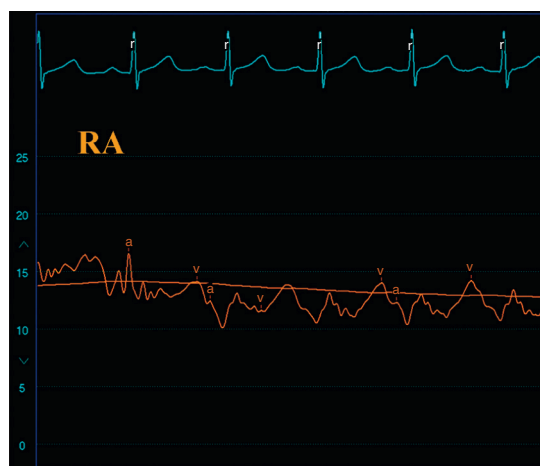


Fig. 13.1 Hemodynamic tracings from a patient with a secundum atrial septal defect. Right atrial pressure tracing is remarkable for a more prominent v wave than a wave, likely caused by tricuspid regurgitation from a dilated right ventricle from long-standing right ventricular volume overload. RA, Right atrial.

EFFECTS OF TREATMENT

The majority of patients in the United States with ASD undergo either surgical or percutaneous closure during childhood. In a study of 123 patients who underwent closure at the Mayo Clinic (Rochester, Minnesota), Murphy et al.¹¹ found that two factors correlated with survival after repair: (1) the age of the patient at the time of the operation ($P < .0001$) and (2) the pulmonary artery systolic pressure ($P < .0027$). Those patients who had a repair prior to age 25 had a long-term survival similar to that of the control population. However, those patients who were older than age 25 and who had a pulmonary artery pressure ≥ 40 mm Hg had a shorter life expectancy than controls. At the 25-year follow-up, survival was 39% versus 74% ($P < .0001$). The increase in mortality was related to the development of congestive heart failure, atrial fibrillation, or cerebrovascular accident. Kobayashi et al.¹² demonstrated that defect closure in patients with Qp/Qs ≤ 3 and a pulmonary artery pressure ≤ 50 mm Hg, or with Qp/Qs ≥ 3 regardless of pulmonary artery pressure, resulted in increased exercise capacity. Gatzoulis et al.¹³ found that as many as 60% of patients with ASD and atrial fibrillation continued to experience this arrhythmia after surgical closure of their defect. Murphy et al.¹¹ reported that of 104 patients in sinus rhythm prior to repair, 80 remained in sinus rhythm 27 to 32 years after the procedure.

Transcatheter closure of secundum ASDs has demonstrated excellent outcomes, with current-era technology having a shorter hospital stay and lower complication rates compared with surgical closure (Fig. 13.2A and B).^{14,15}

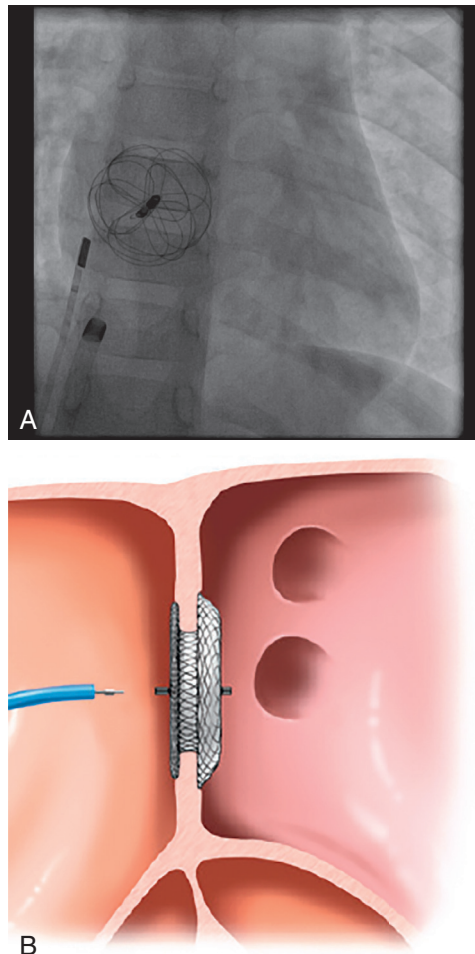


Fig. 13.2 Still frame demonstrating transcatheter closure of a secundum atrial septal defect with (A) a Gore Cardioform device manufactured by Gore & Associates, Newark, DE; and (B) an Amplatzer ASD device. (Amplatzer is a trademark of Abbott or its related companies. Reproduced with permission of Abbott, © 2024. All rights reserved).

VENTRICULAR SEPTAL DEFECT

VSDs are the most common congenital heart lesions, accounting for 20% to 30% of all patients with congenital heart disease. Additionally, this may be an underestimation as small, hemodynamically insignificant VSDs undergo spontaneous closure in the newborn period prior to diagnosis and may be as common as 10% of all pregnancies. This lesion is much less frequently encountered in the adult population given that most undergo spontaneous closure, are surgically repaired, or are hemodynamically insignificant or, if therapy is not undertaken, death occurs in the newborn or childhood period due to cardiac failure. In those who survive to adulthood with significant unrepaired defects, many will have developed pulmonary vascular disease and Eisenmenger syndrome, which is discussed in a later section.

Though intellectually it is easy to simplify the ventricular septum as merely the structure separating the right and left ventricles, it has a complex embryologic origin derived from muscular, endocardial, and conal septal tissues. The intricate development and fusion of each of these tissues during fetal development likely accounts for the high incidence of incomplete separation of the ventricular chambers. Though there are several accepted classification schemes in use for describing the location of these defects, they tend to be generalized into four¹⁶: (1) perimembranous (approximately 75%), found at the crux of the heart, (2) muscular (10%–20%), found within the muscular septum, (3) inlet (approximately 5%), found beneath the tricuspid valve, and (4) outlet (approximately 5%), found in the outflow tract beneath the semilunar valves. The location of the defect tends to have little impact on hemodynamic effect, though it does have an important influence on interventional approach and the likelihood of spontaneous closure, with muscular defects most likely and outlet defects least likely to resolve spontaneously.

HEMODYNAMICS

The hemodynamic effects of VSDs are dependent on the size of the defect and the relative vascular resistances of the pulmonary and systemic circuits, resulting in a left-to-right shunt. These two factors determine the pressure and volume load transmitted from the left to the right ventricle and then pulmonary vasculature. By nature, this increased pulmonary blood flow then increases pulmonary venous return to the left atrium and left ventricle, resulting in a volume load to each of those chambers. Large defects will transmit both a pressure and volume load, while small defects will be pressure restrictive with minimal volume burden.

Additionally, it is important to appreciate the dynamic changes in vascular resistances during the neonatal transitional period. A developing fetus has minimal pulmonary blood flow and a low systemic vascular resistance in part because of placental circulation. Following delivery, an abrupt increase in systemic vascular resistance occurs with the removal of placental circulation. Concurrently, there is an increase in pulmonary blood flow to the newly aerated lungs with a progressive reduction in pulmonary vascular resistance occurring over the next 4 to 6 months. For these reasons, even large VSDs may have a minimal left-to-right shunt in the immediate newborn period, which then increases as the infant matures.

With the continued improvement in noninvasive imaging for diagnostic purposes, purely diagnostic cardiac catheterization for isolated VSDs has become less common unless there is concern regarding significant pulmonary vascular disease. In the current era, transcatheter closure of VSDs has become an important option for select situations, including muscular location, postmyocardial infarction, and some perimembranous lesions.

In patients with large, unrestrictive defects, there is an elevation in right ventricular (Fig. 13.3A) and pulmonary artery systolic pressures equivalent to that of the systemic circulation. Diastolic and mean pressures in the pulmonary arteries will remain lower than systemic until pulmonary vascular disease develops, at which time they will equilibrate. Left atrial and left ventricular end-diastolic pressures will be elevated, reflective of the volume load to those chambers. With a large shunt, there will be a significant increase in oxygen saturations from the right atrium into the right ventricle and pulmonary arteries. The relative ratio of pulmonary to systemic blood flow ($Q_p:Q_s$) tends to be $>2:1$ (Fig. 13.3B).

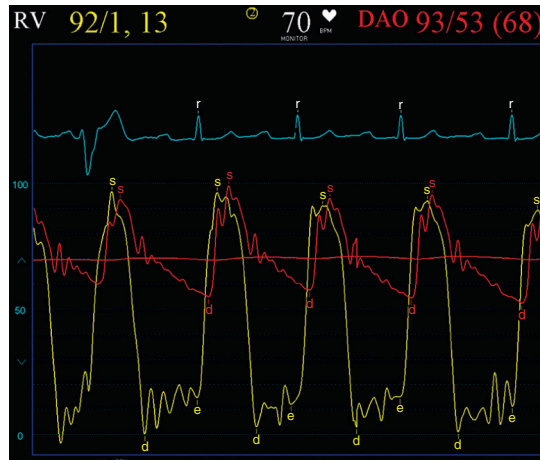
VSDs that are pressure restrictive may still have significant left-to-right shunt flow, as demonstrated in Fig. 13.3C. Right ventricular pressure will be normal, but the pulmonary arteries, the left atrium, and the left ventricle will be dilated from the volume overload.

In patients with smaller, restrictive defects, the right ventricular and pulmonary artery pressures will be proportionately lower, with the left atrial and left ventricular end-diastolic pressure being normal. Additionally, there will be only a modest increase in oxygen saturation from the right atrium into the right ventricle and pulmonary arteries, reflective of minimal volume load. In this situation, $Q_p:Q_s$ is generally $<1.5:1$.

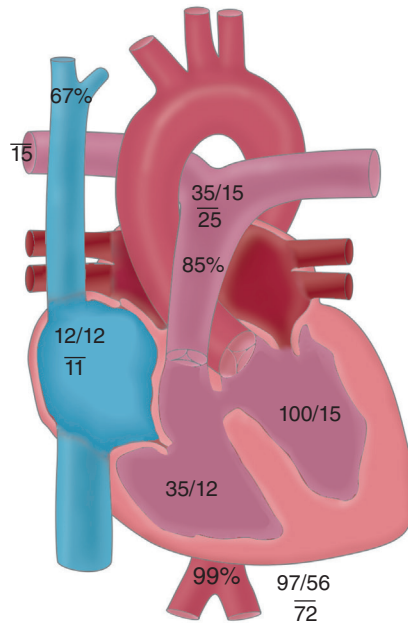
EFFECT OF TREATMENT

As stated previously, the rate of spontaneous closure is dependent on the anatomic location. There is a small, continued risk of endocarditis with persistence of the shunt. Surgical approach is determined by location of the defect, with attempts to avoid a ventriculotomy as this can lead to arrhythmias and aneurysm formation. Inlet and perimembranous defects tend to be approached from a right atriotomy through the tricuspid valve, while outlet defects are approached via pulmonary artery incision through the pulmonary valve. Muscular defects present a surgical challenge, particularly if there are multiple, as they are difficult to visualize in the heavily trabeculated right ventricle. Options here include primary surgical repair, pulmonary artery banding, transcatheter closure, or a combination of procedures.

Among those with hemodynamically significant defects undergoing surgical intervention, outcomes tend to be excellent with minimal mortality even in small infants.¹⁷ Pulmonary artery pressures normalize, and the patients have an accelerated growth after closure.¹⁸



A



B

Fig. 13.3 (A) Simultaneous pressure tracings from the right ventricle and descending aorta. Peak systolic pressures are equal. (B) Schematic of a heart with a restricted ventricular septal defect. The Qp/Qs of this patient prior to closure was approximately 2:1. Note the step-up in pulmonary artery saturation relative to superior vena cava saturation.

Transcatheter closure of muscular VSDs has been utilized since the late 1980s, as described by Lock et al.¹⁹ and has continued to demonstrate good procedural success; however, adverse event rates are higher for this anatomic location.²⁰ Transcatheter closure of perimembranous VSDs remains an option; however, there is a nontrivial incidence of complete heart block associated with this procedure. Although not due to congenital heart disease, postmyocardial infarction VSDs remain a challenging situation with a grim prognosis. Transcatheter closure has been technically successful in the majority of patients, though procedural mortality or need for emergent surgery occurs in approximately 10%.²¹

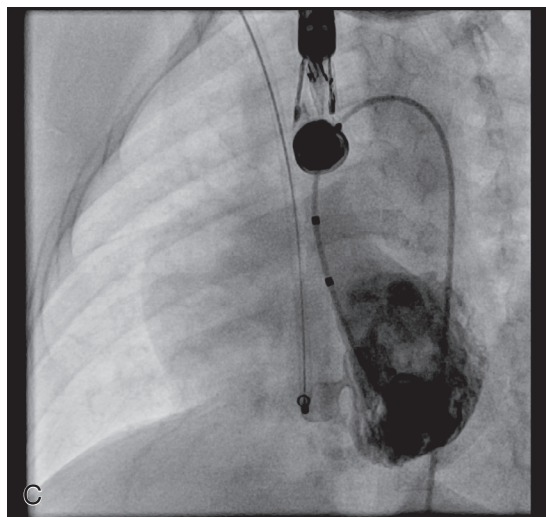


Fig. 13.3, cont'd. (C) Angiogram demonstrating transcatheter delivery of a vascular plug into a muscular ventricular septal defect. *d*, Early diastole; *DAO*, descending aorta; *d*, early diastole; *e*, end diastole; *RV*, right ventricular; *s*, systole.

COARCTATION OF THE AORTA

Coarctation of the aorta refers to a congenital narrowing of the artery occurring in approximately 4% of patients with congenital heart disease. Although it may occur in the transverse aortic arch, the distal thoracic aorta, or even rarely in the abdominal aorta, it occurs most commonly in the proximal descending aorta near the ligamentum arteriosum, the remnant of the ductus arteriosus. The most frequently observed variety consists of a discrete narrowing distal to the left subclavian artery (Fig. 13.4). Oftentimes there is associated distal displacement of the left subclavian artery. The embryologic etiology of aortic coarctation is hypothesized to be an extension of tissue from the ductus arteriosus into the aortic wall, which then constricts during the regression of the ductus in the neonatal period. Most cases of coarctation are sporadic; however, an association with Turner syndrome exists.²² The most commonly reported cardiovascular abnormality coexisting with a coarctation is a bicuspid aortic valve, occurring in 85% of patients. Other abnormalities include intracranial aneurysms in 10% of patients,²³ VSD, aortic medial disease, and abnormalities of the innominate and left subclavian arteries.²⁴

The majority of severe coarctations are discovered and treated during infancy in the United States. When the ductus arteriosus closes in an infant causing a severe obstruction, shock or heart failure develops rapidly. Rapid recognition and administration of prostaglandin to reopen the aortic ampulla alone or the full ductus arteriosus allows for reperfusion of the subthoracic organ systems and recovery of function. Surgical intervention is then undertaken for definitive treatment. Balloon angioplasty and stenting can be considered as rescue therapy in those unresponsive to prostaglandin E or those not deemed surgical candidates in the immediate period. Catheter-based intervention, although rare in infancy, can be utilized as a bridge to definitive surgical repair after organ recovery.

The presentation of a coarctation in adolescence or adulthood is often more subtle and discovered during the evaluation of hypertension. Symptoms are not often reported, though some may experience lower extremity fatigue with exertion and headaches.

Campbell reported that the majority of symptomatic infants with coarctation who were untreated died during the first year of life.²⁵ Reifenshtein et al.²⁶ reported that approximately 20% of asymptomatic children with coarctation survive to adulthood. However, of those surviving into adulthood, 23% died from aortic rupture at an average age of 27 years. Eighty percent of these were rupture of the ascending aorta; the remaining 20% experienced rupture distal to the coarctation. Of the remaining patients, 22% died from infection, including aortitis at an average age of 25 years, and 29% died from intracranial hemorrhage or congestive heart failure.

HEMODYNAMICS

The coarctation represents a fixed resistor to aortic flow with the resulting increase in left ventricular impedance, pressure, and wall stress. The hemodynamic hallmark of coarctation is a pressure gradient across the coarctation, resulting in upper extremity, cerebral, and coronary artery hypertension. This measured gradient may be diminished in the presence of a large patent ductus arteriosus (PDA) in a newborn or with the development of significant aortic collateral

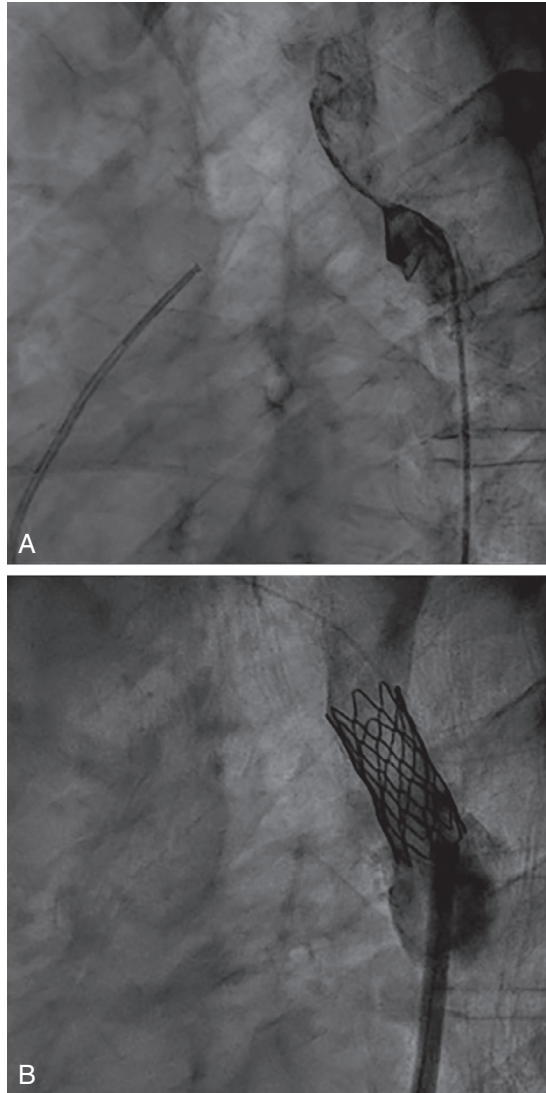


Fig. 13.4 (A) Lateral angiogram from a patient with a severe coarctation of the aorta. Note the discrete narrowing of the aorta to the point of almost being interrupted. (B) Following stenting of the coarctation, there is an interval increase in the vessel caliber.

circulation later in life. Invasively measured gradients (Fig. 13.5) demonstrate a wide pulse pressure proximal to the lesion and a diminished pulse pressure distally. The diminished pulse and the late or delayed pulse wave distal to the coarctation are termed *parvus* and *tardus*, respectively, similar to the pulse described in patients with severe aortic stenosis.

EFFECTS OF TREATMENT

Surgical treatment options for aortic coarctation have included (1) coarctation excision with end-to-end anastomosis, (2) interposition graft, (3) patch aortoplasty, (4) extended end-to-end anastomosis, (5) bypass jump grafting, (6) subclavian flap aortoplasty, and (7) aortic arch advancement. Contemporary treatment includes percutaneous balloon angioplasty and stent implantation,^{27,28} with some studies suggesting primary stent implantation has superior results with fewer complications at mid-term follow-up.²⁹

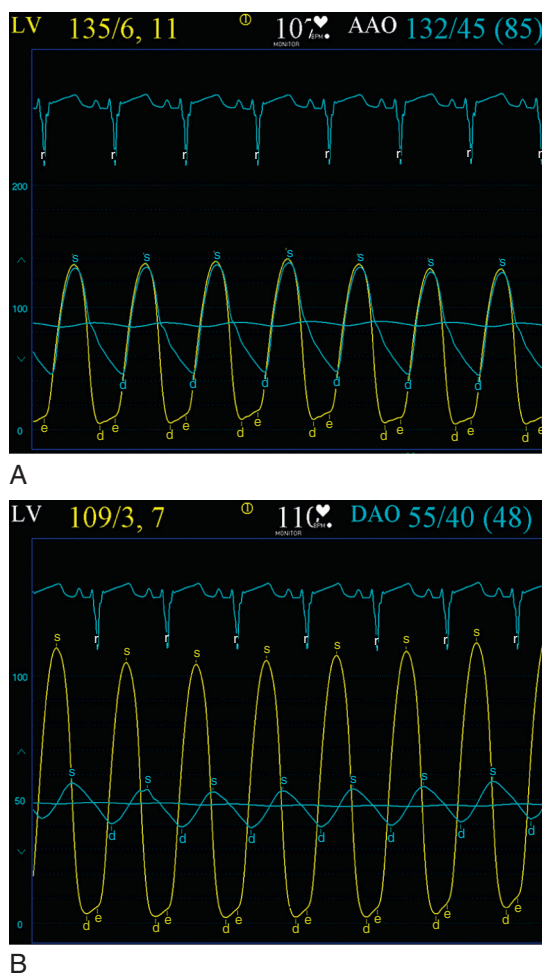


Fig. 13.5 Hemodynamic tracings from a patient with coarctation of the aorta. (A) Simultaneous pressure tracings from the left ventricle and the ascending aorta. Peak systolic pressures are equal, demonstrating no valvular aortic stenosis. However, the pulse pressure of the ascending aorta is wide due to aortic valve regurgitation. (B) Simultaneous pressure tracings from the left ventricle and the descending aorta distal to the coarctation. A significant drop occurs in systolic pressure distal to the coarctation. The lower pulse pressure and systolic delay in the descending aorta represent the “parvus et tardus” of coarctation. AAO, Ascending aorta; DAO, descending aorta; e, end diastole; LV, left ventricle; s, systole.

Unfortunately, after successful surgical or percutaneous therapy for coarctation, mid-term and late-term complications may still occur. Cohen et al.³⁰ reported that even after coarctation repair the mortality of these patients is elevated. In an analysis of 571 patients with repaired coarctations, the mean age of death was 38 years, with 37% dying from premature coronary artery disease and earlier repair associated with improved survival. Additionally, certain types of repairs place patients at increased risk for aortic aneurysm formation and rupture. Mendelsohn et al.³¹ reported the incidence of aneurysm development at the site of patch repair to be as high as 30%. Persistent systemic hypertension is not uncommon for many patients with repaired coarctation, even without anatomic obstruction. O'Sullivan et al.³² demonstrated systemic hypertension in 20% to 25% of children and adolescents repaired as infants or young children with no identifiable aortic arch obstruction. Similarly, of those who undergo repair after age 40, 50% will continue to experience hypertension at rest, and a substantial fraction will have exercise-induced hypertension.³³ This persistent hypertension may be due, in part, to impaired aortic distensibility but also to intrinsic abnormalities of the microvasculature, baroreceptors, and neurohumoral regulation.^{34–36}

EBSTEIN ANOMALY

Ebstein anomaly is an eponym used to designate a syndrome in which the tricuspid valve is apically displaced with variable degrees of tricuspid regurgitation. It has an incidence of <1% of all patients with congenital heart disease. Characteristically, the septal and posterolateral leaflets are positioned below the atrioventricular groove and tend to be deformed and variably adherent to the ventricular septum.³⁷ Coaptation of the tricuspid leaflets is commonly abnormal, resulting in important tricuspid regurgitation. Apical displacement of the tricuspid valve leads to a right ventricle becoming partially "atrialized." This atrialized portion encroaches on the right ventricle, resulting in a relatively small ventricular chamber with impaired compliance. Pulmonary blood flow may additionally be impaired by true anatomic pulmonary atresia or by functional right ventricular outflow tract obstruction as a result of severe tricuspid insufficiency, leading to the inability of the right ventricle to generate sufficient pressure to open the pulmonary valve.

The majority of patients with Ebstein anomaly have an interatrial communication, typically in the form of a stretched patent foramen ovale or a secundum ASD. As with other atrial communications, flow through this defect is determined by the relative compliance of the right and left ventricles. With Ebstein anomaly, this commonly results in a right-to-left shunt and systemic desaturation. Additionally, atrial arrhythmias are encountered in 10% to 25% of patients with Ebstein anomaly and are frequent causes of morbidity.^{38,39}

HEMODYNAMICS

The variability of hemodynamics in this condition is predicated on the severity of tricuspid regurgitation and the systolic and diastolic function of the right ventricle. In mild cases of Ebstein anomaly, the only hemodynamic observation may be that of tricuspid regurgitation manifested by a dominant v wave in the right atrium, with modest global elevation of right atrial pressures. As in other situations with atrial shunts, flow through the defect is determined by the relative compliance of the right and left ventricles. With mild disease, right ventricular compliance remains below that of the left ventricle, resulting in left-to-right shunting at the atrial level. In more severe cases, right-to-left shunts may occur across the ASD due to the chronic volume overload and loss of compliance of the right ventricle, thus leading to systemic hypoxemia (Fig. 13.6).

Neonates presenting with severe disease represent a particularly challenging population, demonstrating severely elevated central venous pressure, massive cardiomegaly, and cyanosis secondary to a combination of right-to-left atrial shunting, low systemic venous saturation, and pulmonary venous desaturation reflective of ventilation/perfusion (V/Q) mismatching. Efforts are made to augment pulmonary blood flow through the use of pulmonary vasodilators and prostaglandins in addition to inspired oxygen to increase pulmonary venous saturations during the neonatal transitional period.

Special mention of a unique circulation, termed *circular shunt*, is merited at this time. This situation is a result of Ebstein anomaly (or any condition with significant tricuspid regurgitation and an atrial communication) with the addition of pulmonary regurgitation and an aorto-pulmonary connection, typically a PDA. Here, an attempt is made to augment pulmonary blood flow from the aorta across the aorto-pulmonary shunt. However, rather than flow being directed to the pulmonary capillary bed for gas exchange, it is directed across the incompetent pulmonary valve into the right ventricle with subsequent regurgitation across the dysfunctional tricuspid valve. From here, right-to-left shunting across the atrial communication leads to the ejection of further desaturated blood into the systemic circulation, causing a downward, "circular" spiral of worsening hypoxemia.

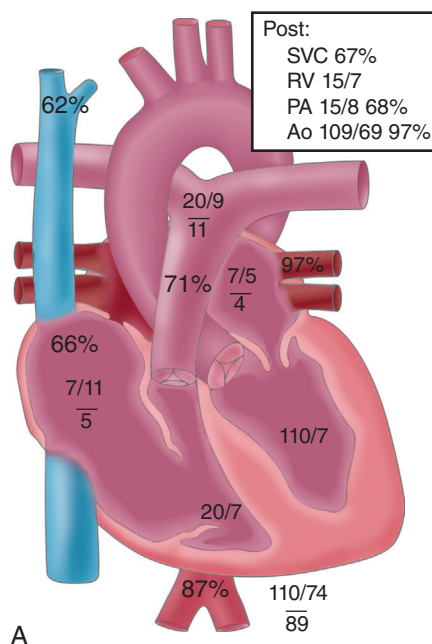
EFFECTS OF TREATMENT

Freud et al.⁴⁰ found that for neonates diagnosed in utero there are generally less than ideal outcomes, with approximately 30% mortality prior to hospital discharge regardless of treatment strategy. In their study, only half of those surviving had a biventricular circulation.

Those reaching or diagnosed in adulthood have a more benign course with a reasonable duration of life, though they commonly need intervention once symptomatic.⁴¹ For adults with heart failure or tachyarrhythmias refractory to medical therapy, closure of the interatrial communication and repair or replacement of the tricuspid valve are recommended.^{42,43} Danielson et al.⁴⁴ have reported excellent results with surgical intervention, including tricuspid repair, in this patient population. Robust, long-term series of functional status are lacking in the literature; however, there have been reports of survival to 85 years.⁴⁵

TETRALOGY OF FALLOT

Tetralogy of Fallot is a syndrome composed of the tetrad of (1) an unrestricted malalignment VSD, (2) right ventricular outflow tract obstruction, (3) right ventricular hypertrophy, and (4) an overriding aorta. Though the original description of the lesion was first made in 1673 by Danish monk Nicholas Steno,⁴⁶ the unifying features of tetralogy were published by Arthur Louis Etienne Fallot⁴⁷ in 1888, thus the well-known eponym. The incidence of tetralogy of Fallot is approximately 3.5% of infants born with congenital heart disease.⁴⁸ Embryologically, the hallmark of the lesion is anterior-superior displacement of the conal septum, creating the VSD, the overriding aorta, and the right ventricular outflow tract obstruction with subsequent right ventricular hypertrophy.



A



B

Fig. 13.6 Hemodynamic tracings from a patient with unrepaired Ebstein anomaly of the tricuspid valve and an atrial septal defect with cyanosis. (A) Schematic of the patient's heart with Ebstein anomaly. Note the apical displacement of the tricuspid valve. Bidirectional shunting exists at the atrial septum as demonstrated by the oximetry step-up from the superior vena cava to the pulmonary artery and step-down from the pulmonary vein to the aortic saturation. Note that in this patient an atrial septal defect occlusion device was placed, resulting in the resolution of the bidirectional atrial shunting, a fall in right ventricular and pulmonary artery pressures, an improvement in systemic saturation, and the resolution of cyanosis (*box inset*). (B) Right atrial pressure tracing showing elevated v waves consistent with tricuspid regurgitation. *a*, a wave; *v*, v wave.

Abnormalities associated with this lesion include right aortic arch (25%),^{49,50} ASD (10%),⁴⁹ and coronary artery anomalies (10%).⁵¹ In approximately 5% of cases, there is a major coronary artery coursing over the infundibular free wall, altering surgical options.⁵²

Clinical features of the lesion are directly related to the degree of right ventricular outflow tract obstruction. Those patients with a moderate-to-severe degree of obstruction will demonstrate a reduction in pulmonary blood flow and net



Fig. 13.6, cont'd (C) Right ventricular pressure tracing showing normal, low pressures. *Ao*, Aorta; *PA*, pulmonary artery; *RA*, right atrium; *RV*, right ventricle; *SVC*, superior vena cava.

right-to-left flow at the VSD, leading to systemic desaturation (termed *blue tet*). Conversely, those with minimal to no degree of obstruction to pulmonary blood flow (termed *pink tet*) will present analogous to an isolated VSD and have a net left-to-right shunt, resulting in increased pulmonary blood flow and normal systemic saturations.

Occasionally, unrepaired patients will present with profound desaturation episodes, termed *tet spells* or *hypercyanotic spells*, which can progress to seizure, syncope, and death.⁵³ Though the precise mechanism for these episodes remains elusive, it likely represents an interplay of factors including a reduction in systemic venous return, dynamic muscular right ventricular outflow tract obstruction, increased pulmonary vascular resistance, and a reduction in systemic vascular resistance. These all consequently lead to a reduction in pulmonary blood flow and augmentation of the right-to-left ventricular shunt. Therapy is aimed at these factors and includes increasing intravascular volume, oxygen supplementation, reduction of inotropic stimulation (morphine, beta-blockade), and administration of systemic vasoconstrictors (α 1-agonist). Occasionally, a patient will need to be sedated or paralyzed, requiring intubation, to break a hypercyanotic spell.

Left unrepaired, the natural history of tetralogy of Fallot is poor, with most patients succumbing as children. Significant morbidity is due to the effects of the right-to-left shunt and includes cerebral abscess, stroke, endocarditis, and the negative effects of erythrocytosis. Bertranou et al.⁵⁴ reviewed a large volume of published autopsy cases and reported that the untreated survival of tetralogy of Fallot is 66% at 1 year, 49% at 3 years, 24% at 10 years, 11% at 20 years, 6% at 30 years, and 3% at 40 years.

HEMODYNAMICS

The hemodynamics of unrepaired tetralogy of Fallot can be separated into those with moderate-to-severe right ventricular outflow tract obstruction and those with minimal to no obstruction. Similarities between the groups include a normal to mildly elevated right atrial pressure reflective of the degree of right ventricular hypertrophy and subsequent change in ventricular compliance. Right ventricular and left ventricular pressures are equivalent, commensurate with the large, pressure unrestrictive VSD, though there are rare reports of the septal defect being pressure restrictive.⁵⁵ With a progressive increase in right ventricular outflow tract obstruction, regardless of the anatomic level, impedance to pulmonary blood flow increases with subsequent reduction in pulmonary artery pressures. Eventually, pulmonary impedance becomes greater than systemic, and the direction of flow across the VSD becomes predominantly right to left. Progressive hypoxemia tends to ensue and generally harkens the need for interventional therapy, either in the operating room or catheterization laboratory.

The hemodynamics of patients who have had prior repair with the closure of the VSD and augmentation of the right ventricular outflow tract are generally reflective of the function of the ventricle to pulmonary connection. Residual obstruction to pulmonary blood flow leads to a pressure load on the right ventricle with an increased demand for systolic pressure generation. Significant regurgitation across an incompetent, surgically augmented right ventricular outflow tract leads to a reduction in diastolic pulmonary artery pressures (ventricularization of pulmonary artery pressure) and a volume load on the right ventricle (Figs. 13.7–13.9). Depending on the chronicity and severity of these residual

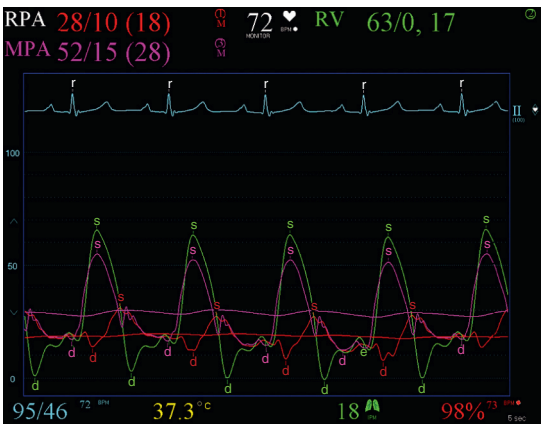


Fig. 13.7 Hemodynamic tracings from a patient with repaired tetralogy of Fallot. This patient underwent a transannular patch of the pulmonary valve and infundibulum and was left with free pulmonary regurgitation. The right ventricular main pulmonary artery and right pulmonary artery tracings are shown. The right ventricular pressure is elevated but with a minimal gradient across the right ventricular outflow tract. However, branch pulmonary stenosis exists with a lower right pulmonary artery pressure. The waveform of the pulmonary artery tracing is notable for a ventricular morphology, that is, no diastolic notch is present from pulmonary valve closure, and the end-diastolic pressures of the pulmonary artery and right ventricle are essentially equal. This is consistent with severe pulmonary insufficiency. *d*, Early diastole; *e*, end diastole; *MPA*, main pulmonary artery; *RPA*, right pulmonary artery; *RV*, right ventricle; *s*, systole.

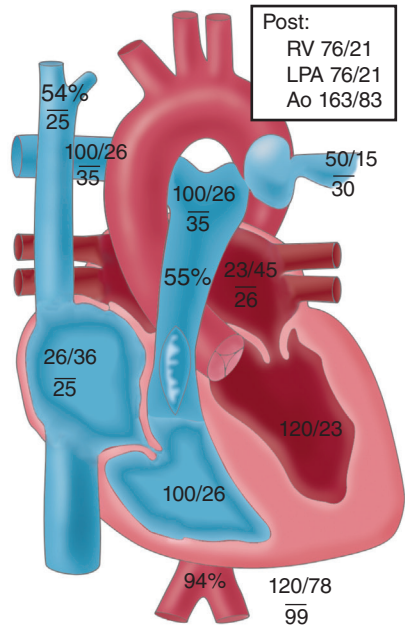


Fig. 13.8 Schematic of a heart from a patient with repaired tetralogy of Fallot. The left pulmonary artery is stenotic from a prior Potts shunt, resulting in pulmonary hypertension of the right pulmonary artery. The hemodynamics after placement of a covered stent into the left pulmonary artery, eliminating both the stenosis and aneurysm, are shown (*box inset*). *Ao*, Aorta; *LPA*, left pulmonary artery; *RV*, right ventricle.

lesions, right ventricular systolic and diastolic dysfunction ensue. This is subsequently exhibited as elevation in right ventricular end-diastolic and right atrial pressures and a reduction in baseline cardiac output and its ability to augment during times of increased demand. The interconnectedness between the ventricles is also demonstrated by a subsequent reduction in left ventricular function.

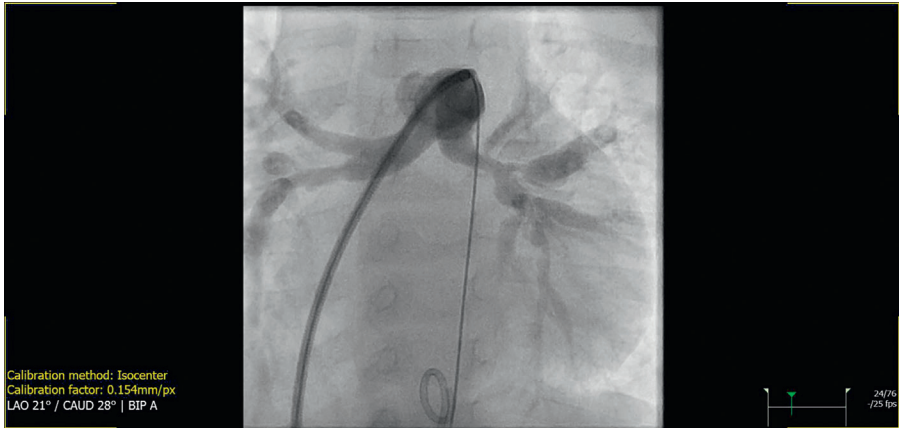


Fig. 13.9 Pulmonary artery angiogram from another patient with tetralogy of Fallot and pulmonary atresia s/p (status post) repair with a right ventricle-pulmonary artery conduit, leaving the patient with left pulmonary artery stenosis. The patient had a hypertensive right pulmonary artery and nonpulsatile distal left pulmonary artery pressure, with elevated right ventricular pressures.

The hemodynamics of pregnancy in patients with unrepaired tetralogy of Fallot deserves special attention. During pregnancy, systemic vascular resistance decreases, increasing the degree of right-to-left shunting and worsening hypoxemia. Additionally, those who have had earlier surgical repair demonstrate a subsequent acceleration in right ventricular remodeling and dilation⁵⁶ and may benefit from intervention before conception to optimize hemodynamic performance in anticipation of the cardiovascular stressors of pregnancy. Therefore it is imperative that all patients with tetralogy of Fallot receive counseling from an adult congenital heart disease physician prior to conception.⁵⁷

THERAPY

While surgical correction of those patients with minimal obstruction to pulmonary blood flow is essentially closure of the ventricular defect, intervention in those with reduced pulmonary blood flow carries several options. Historically, tetralogy of Fallot is of interest as it was the first cyanotic congenital heart defect to receive palliation in 1945 and therefore represents the archetypal reference for other cyanotic lesions. Here, augmentation of pulmonary blood flow was achieved by the creation of a systemic-to-pulmonary artery shunt via the right subclavian artery. This procedure is termed the *Blalock-Taussig shunt*,⁵⁸ though Vivien Thomas also played a critical role in its development and many feel that this shunt should be called *Blalock-Taussig-Thomas (BTT)*. As this procedure interrupts normal systemic flow to that upper extremity, the technique was later modified to use a tube graft connection between the subclavian and pulmonary arteries. Other options for systemic-to-pulmonary connections included the Potts (descending aorta to left pulmonary artery) and Waterston (ascending aorta to right pulmonary artery) shunts. These had the disadvantages of pulmonary artery distortion and difficulty controlling the pressure transmission to the pulmonary bed, resulting in vascular disease.

In the current era surgical therapy includes closure of the ventricular defect and relieving obstruction to pulmonary blood flow, be it subvalvular or supra-valvular. While complete relief of outflow tract obstruction with a transannular patch was previously thought optimal,^{59–61} creation of significant pulmonary insufficiency and its mid-term to long-term hemodynamic sequelae have been questioned. As discussed previously, significant pulmonary insufficiency creates a volume load on the right ventricle with subsequent hemodynamic derangements, including systolic and diastolic dysfunction, elevation in central venous pressure, and associated atrial and ventricular arrhythmia.⁶² This has led to some centers performing a “valve-sparing” repair when possible in an attempt to preserve the function of the pulmonary valve with the acceptance of a modest degree of stenosis.⁶³

Timing of intervention to address pulmonary insufficiency continues to be a fluid discussion as diagnostic and interventional techniques evolve.^{64–67} Ideally intervention will take place prior to the development of irreversible clinical or subclinical dysfunction while minimizing the total number of procedures required over a patient's lifetime. While historically surgical pulmonary valve replacement has been standard for addressing this issue, transcatheter pulmonary valve replacement has shown promise, though long-term data on durability is currently an active area of interest.⁶⁸

PERIPHERAL PULMONARY ARTERY STENOSIS

Peripheral pulmonary artery stenosis is a condition where obstructive lesions occur at or distal to the right- and left-branch pulmonary arteries. Though they can rarely occur in isolation, the disease is more frequently encountered in

postoperative patients (tetralogy of Fallot, dextro-transposition of the great arteries [d-TGA], truncus arteriosus) (Fig. 13.10), those with associated syndromes (Noonan, LEOPARD, William, Alagille), or occasionally with systemic vasculitis.⁶⁹ Though the majority of these patients are diagnosed in childhood, occasionally an adult patient will present with a presumptive diagnosis of pulmonary embolism following an abnormal ventilation-perfusion scan. Subsequent angiographic imaging demonstrates the multiple obstructive lesions, distinguishing the processes. Progressive dyspnea and fatigue are severe and universal presenting signs in the adult population.⁷⁰

HEMODYNAMICS

As with other obstructive lesions, peripheral pulmonary artery stenosis reduces vascular cross-sectional area, increasing ventricular impedance, resulting in a progressive pressure load on the corresponding ventricle. The pulmonary tree, however, presents a unique situation of a parallel circulation as the multiple arborizing branch points tend to redistribute flow to less affected vascular segments. This results in variable areas of overperfusion and underperfusion with segments unaffected by stenosis absorbing the increased flow. The right ventricular pressure generated will then become a function of the summed cross-sectional area of the stenotic and unaffected vessels in the pulmonary tree. As the total cross-sectional area of the vascular tree is progressively reduced by the stenotic regions, impedance to flow



Fig. 13.10 Hemodynamic tracings from a patient with left and right pulmonary artery stenosis after an arterial switch operation for *d*-type transposition of the great arteries. (A) The main pulmonary artery pressure is markedly elevated relative to the right pulmonary artery pressure distal to the site of stenosis. (B) The main pulmonary artery pressure is markedly elevated relative to the left pulmonary artery pressure distal to the site of stenosis. Note the appearance of the main pulmonary artery pressure waveform. It appears like a ventricular waveform because of severe pulmonary insufficiency. *DAO*, Descending aorta; *e*, end diastole; *LPA*, left pulmonary artery; *MPA*, main pulmonary artery; *RPA*, right pulmonary artery; *s*, systole.

increases with a concurrent increase in the right ventricular pressure. This pressure load will then be transmitted to the vascular beds not protected by proximal stenosis, potentially resulting in endothelial damage and vascular disease in the previously normal vessels. At its extreme, this situation leads to right ventricular hypertension and failure with a patchwork of variably affected segments of lung, with normal distal pulmonary artery pressures in those vessels having proximal stenosis and pulmonary vascular disease in those vessels unaffected by obstruction (Fig. 13.11).

Indications for therapy include contralateral pulmonary arterial hypertension, right ventricular hypertension, or significant flow imbalance, particularly in younger patients with continued growth potential of the lungs.

EFFECTS OF TREATMENT

Treatment options for peripheral pulmonary stenosis include both surgical and transcatheter intervention, though if the lesion is distal to the hilum, percutaneous therapy tends to be the treatment of choice. Transcatheter low-pressure and high-pressure angioplasty, cutting balloons, and stent implantation have all been acutely successful depending on the lesion characteristics.⁷¹

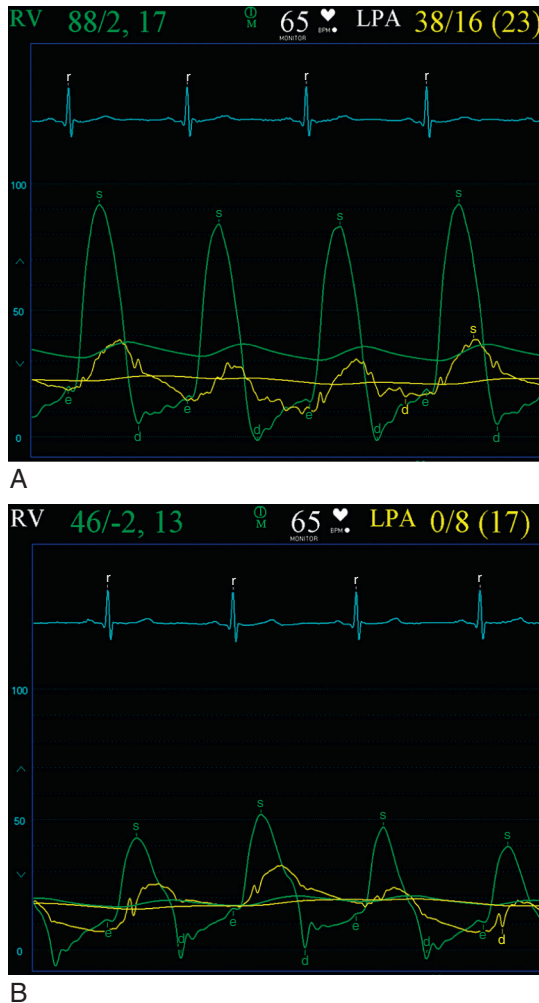


Fig. 13.11 Hemodynamic tracings from a patient with left pulmonary artery stenosis and free pulmonary regurgitation. (A) The right ventricular pressure is markedly elevated (80% systemic), and equilibration of diastolic pressures between the left pulmonary artery and the right ventricle is consistent with severe regurgitation. (B) After balloon dilatation and stent implantation across the left pulmonary artery stenosis, the right ventricular pressure has dropped to 41% of systemic. e, End diastole; LPA, left pulmonary artery; RV, right ventricle; s, systole.

Due to the low prevalence of this disease process in the adult population, large volume studies are lacking, making robust outcome conclusions challenging. Kreutzer et al.⁷⁰ have the largest series to date with 12 patients without prior cardiac surgery and no evidence of vasculitis undergoing hemodynamic evaluation. One patient was found to have severely elevated distal pulmonary artery pressures, did not undergo intervention, and was listed for lung transplantation. Of the remaining 11, all underwent balloon angioplasty of stenotic lesions. One patient expired post-procedure because of pulmonary hemorrhage and another had lesions unresponsive to angioplasty, continued to experience symptoms, and died while awaiting transplant. The remaining nine patients had improved symptoms sustained for 52 ± 32 months.

In 2003 Rothman et al.⁷² added their experience of four adult patients to the literature. This group demonstrated the feasibility of combined angioplasty and stenting in multiple lobar vessels with acutely improved symptoms in all patients. Unfortunately, the lesions had a high recurrence rate within several months of intervention, necessitating re-intervention or lung transplantation.

EISENMENGER SYNDROME

Eisenmenger syndrome was first described by Victor Eisenmenger in 1897 when he published a case report of a cyanotic gentleman with signs of right heart failure and a murmur consistent with a VSD, which was confirmed by autopsy.⁷³ It is characterized by a dynamic and multifactorial process of pulmonary vascular remodeling in the setting of a left-to-right shunt. This shunt causes endothelial damage leading to histopathologic intimal proliferation, smooth muscle propagation, medial hypertrophy, plexiform lesions, and thrombosis. Initially, these cellular changes can be reversible with closure of the shunt. However, given a variable time frame depending on the lesion characteristics and intrinsic factors of the subject, these changes become irreversible and result in the reversal of the shunt from right to left and subsequent systemic hypoxemia. This clinical complex can now be appropriately termed *Eisenmenger syndrome* (Fig. 13.12).⁷⁴

While the majority of lesions leading to Eisenmenger syndrome are congenital in nature (ASD, VSD, PDA, atrio-ventricular canal, etc.), there are also palliative surgical shunts used to augment pulmonary blood flow (Potts and Waterston), which may result in excessive pulmonary blood flow and endothelial damage.

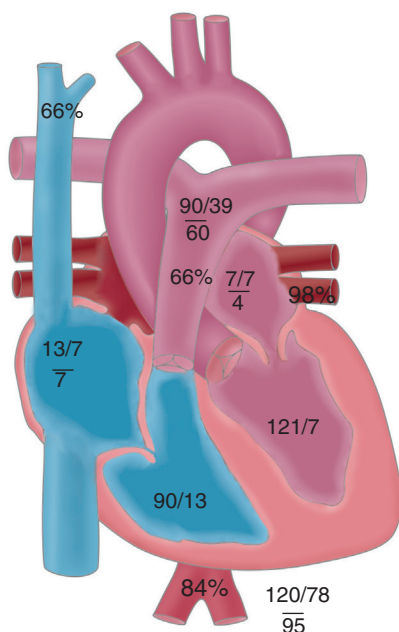


Fig. 13.12 Schematic of a heart from a patient with Eisenmenger physiology from a long-standing atrial septal defect. Chronic right ventricular volume overload from left-to-right shunting at the atrial septal defect has led to a severely dilated right ventricle with elevated end-diastolic pressures from decreased compliance. As shunting at the atrial septal defect is determined by the relative compliance of the two ventricles, the direction of flow across the shunt then became right to left, as shown by the step-down in oxygen saturations from the pulmonary vein to the aorta. Note the severely elevated pulmonary and right ventricular systolic pressures from pulmonary vascular occlusive disease.

Clinically, patients with Eisenmenger syndrome have multisystem organ dysfunction resulting in chronic hypoxemia, bleeding and thrombotic episodes, infectious predilection, arrhythmias, and sudden death.⁷⁵ Chronic cyanosis causes an elevation in erythropoietin and secondary erythrocytosis, which can lead to iron deficiency, headaches, dizziness, and paresthesias.⁷⁶ In a review of the literature from 1966 to 1998, Vongpatanasin et al.⁷⁷ reported that at 10 years after diagnosis 80% of patients survived, at 15 years 77% survived, and at 25 years only 42% survived. Saha et al.⁷⁸ found that 30% of patients die of sudden death, 25% die of congestive heart failure, and 15% die of pulmonary hemorrhage.

HEMODYNAMICS

Cardiac catheterization is essential in the evaluation of all patients with pulmonary hypertension, including those with congenital heart defects. As there are a variety of congenital heart defects leading to pulmonary vascular disease, the hemodynamics will be a spectrum reflective of the severity of vascular disease and residual shunting through the defect. Calculation of flows to the pulmonary and arterial circuits is made using directly measured oxygen saturations from the systemic artery, pulmonary artery, pulmonary veins, and systemic vein. The ratio of those flows can then be presented as a Qp:Qs ratio. In those with Eisenmenger syndrome, pulmonary blood flow will be reduced due to right-to-left shunting and therefore Qp:Qs will be <1.

Right atrial pressures will be normal in those with early disease and elevated in those with right ventricular diastolic dysfunction. Right ventricular pressures will be elevated and reflective of any residual shunts and degree of pulmonary pressures. Pulmonary artery pressures will be elevated, again dependent on residual shunts and degree of vascular disease.

Pulmonary resistances are then calculated by dividing the change in mean pressure over the pulmonary vascular bed by the total flow through that circuit. Administration of pulmonary vasodilator medication can be added and evaluated for changes in pulmonary pressures and resistances. In those with Eisenmenger syndrome, pulmonary resistances will be equivalent to or greater than that of the systemic circulation and tend to be poorly responsive to vasodilator medication.⁷⁹

EFFECTS OF TREATMENT

Therapeutic options have traditionally been palliative in nature, though there have been some recent advances in pharmacologic management. Avoidance of dehydration, iron supplementation to treat iron-deficiency anemia, supplemental oxygen, anticoagulation, and antiarrhythmic therapy have all been used for variable efficacy.⁸⁰ Phlebotomy with isotonic volume replacement is now rarely performed as this promotes iron deficiency and is reserved for those with overt symptoms of hyperviscosity who have been adequately treated with iron supplementation. A variety of classes of pulmonary vasodilators including prostacyclin analogs, phosphodiesterase inhibitors, and endothelin-receptor antagonists have been employed with modest success. End-stage treatment for Eisenmenger syndrome involves lung transplantation with the repair of the underlying defect or combined heart-lung transplantation. Survival for adults with lung transplantation and cardiac repair is 70% and under 50% at 1 and 4 years, respectively. Combined heart-lung transplantation has a survival rate of 60% and under 30% at 1 and at 10 years, respectively.⁸¹ Pregnancy is strongly discouraged and is generally considered contraindicated because of the high mortality rates for mother and fetus, both prepartum and postpartum.

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